CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Third Semester of B.Sc.(Physics) Examination March 2018
PD 201 Mathematical Physics-II

Date: 26/03/2018, Monday **Time**: 01:30 p.m. to 02:00 p.m. **Maximum Marks**: 20

MCQ

Important Instructions:

- Provide all necessary information as required in the answer sheet.
- Attempt all questions.
- Answer the questions in the question paper itself.
- Use of cell phones is strictly prohibited.
- Use of non-programmable calculators may be allowed if needed.

Q.1. Choose the correct answer for the following questions. [20]

1. Which of the following is two dimensional heat equation?

   a. $\frac{\partial^2 z}{\partial t^2} = c^2\left(\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2}\right)$

   b. $\frac{\partial z}{\partial t} = c^2\left(\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2}\right)$

   c. $\frac{\partial z}{\partial t} = c^2\left(\frac{\partial^2 z}{\partial x^2}\right)$

   d. $\frac{\partial^2 z}{\partial x^2} = c^2\left(\frac{\partial^2 z}{\partial x^2}\right)$,

   where c is constant.

2. A partial differential equation contain __________.

   a. one independent variable and more than one dependent variable
   b. one independent variable and one dependent variable
   c. one dependent variable and more than one independent variable
   d. more than one independent variable and more than one dependent variable.

3. For solving partial differential equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$, the number of initial and boundary conditions required are __________ respectively.

   a. 2 and 2 b. 2 and 1 c. 1 and 2 d. 1 and 3

4. Which of the following possible form of series solution for the differential equation $x(x-1)(x-2)y'' - 5y' + 17y = 0$?

   a. $\sum\limits_{n=0}^{\infty} a_n x^n$
   b. $\sum\limits_{n=0}^{\infty} a_n (x-1)^n$
   c. $\sum\limits_{n=0}^{\infty} a_n (x-2)^n$
   d. $\sum\limits_{n=0}^{\infty} a_n (x-5)^n$,

   for some a_n.

5. Which of the following point is the regular singular point of the differential equation
   $x^4(x-2)^3(x-7)y'' - (x-5)(x-2)y' + 5xy = 0$?

   a. 0 b. 2 c. 5 d. 7

6. For Legendre polynomial $P_n(x)$, $\int\limits_{-1}^{1} (P_2(x))^2 \, dx$=__________.

   a. $\frac{2}{3}$ b. $\frac{2}{5}$ c. 1 d. 0

7. For Legendre polynomial $P_n(x)$, $P_1(x)$=__________.

   a. 1 b. x c. x^2 d. $\frac{x}{2}$

8. For Legendre polynomial $P_n(x)$, which of the following is true?

   a. $\sqrt{1-2xz+z^2} = \sum\limits_{n=0}^{\infty} z^n P_n(x)$
   b. $\frac{1}{(1-2xz+z^2)^2} = \sum\limits_{n=0}^{\infty} z^n P_n(x)$
   c. $\frac{1}{1-2xz+z^2} = \sum\limits_{n=0}^{\infty} z^n P_n(x)$
   d. $\frac{1}{\sqrt{1-2xz+z^2}} = \sum\limits_{n=0}^{\infty} z^n P_n(x)$

9. The particular solution of the differential equation $x^2\frac{d^2y}{dx^2}+x\frac{dy}{dx}+(x^2-16)y = 0$ is called__________.
   a. Bessel's function of order 16
   b. Bessel's function of order 4
   c. Bessel's function of order 2
   d. Bessel's function of order 1

10. For Bessel function $J_n(x)$, $J_0'(x)$=__________.

a. $J_1(x)$ b. $-J_1(x)$ c. $J_2(x)$ d. $-J_2(x)$

11. For a function $z = f(x, y)$, the error in z computed as__________.

a. $dz = (\frac{\partial f}{\partial x})\, dx + (\frac{\partial f}{\partial y})\, dy$
b. $dz = (\frac{\partial f}{\partial x})\, dx - (\frac{\partial f}{\partial y})\, dy$
c. $dz = (\frac{\partial f}{\partial x})\, dx + (\frac{\partial f}{\partial y})^2\, dy$
d. $dz = (\frac{\partial f}{\partial x})^2\, dx + (\frac{\partial f}{\partial y})\, dy$

12. If y is the true value of a number and y' is the approximate value obtained by computation, then $\frac{|y-y'|}{y} \times 100$ is called __________.

a. absolute error
b. relative error
c. percentage error
d. none of these

13. Which of the following function is even?

a. $x^3 \cos 2x$
b. $x^3 \sin 2x$
c. e^{2x}
d. $x^2 - 5$

14. $\frac{\Gamma(n+1)}{\Gamma(n)}$=__________.

a. $n - 1$ b. n c. $n!$ d. $(n - 1)!$

15. B(0.5,0.5)=__________.

a. π b. $\frac{\pi}{2}$ c. 2π d. $2\sqrt{\pi}$

16. For error function $erf(x)$, which of the following statement is true?

a. $erf(x) = erfc(x) - 1$
b. $erfc(x) = \frac{2}{\sqrt{\pi}} \int\limits_0^\infty e^{-t^2}\, dt$
c. $erf(0) = 1$
d. $erfc(x) + erf(x) = 1$

17. The error function is __________.
a. an odd function
b. an even function
c. neither odd function nor even function
d. exponential function.

18. Let $f(t) = \begin{cases} -\pi, & \text{for } -\pi < t < 0, \\ t, & \text{for } 0 < t < \pi. \end{cases}$ Then at $t = 0$, the Fourier series of $f(t)$ converges to __________.

a. $\frac{f(0+)+f(0-)}{2}$
b. $\frac{f(0+)-f(0-)}{2}$
c. $f(0+)$
d. $f(0-)$,

where $f(0+)$ and $f(0-)$ are right hand left and left hand limit of function $f(t)$ at 0 respectively.

19. Let $f(x) = x^{11}$ be the function defined on $[-5, 5]$. Then Fourier series of $f(x)$ contains__________.

a. only sine terms
b. only cosine terms
c. only constant terms
d. none of above

20. Which of the following condition is not necessary for existence of Fourier series of function on some interval?

a. function is single valued
b. function is bounded
c. function is continuous
d. function has finite number maxima and minima

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
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PD 201 Mathematical Physics-II

Date: 26/03/2018, Monday **Time**: 02:00 p.m. to 4:30 p.m. **Maximum Marks**: 50

Instructions:

1. **Section I and II must be attempted in seperate answer sheets.**
2. **Make suitable assumptions and draw neat figures wherever required.**
3. **Use of non-programmable calculator is allowed.**
4. **Show necessary calculations.**
5. **Figures to right indicate marks.**

SECTION-I

Q-2 Answer the following questions as directed. [20]

(a) Write the general form of three dimensional Laplace's equation and two dimensional wave equation. [02]

(b) The period of a simple pendulum is $T = 2\pi\sqrt{\frac{l}{g}}$, find the maximum error in T due to the possible error upto 1% in l and 2.5% in g. [02]

(c) Define error function $erf(x)$. Expand $erf(x)$ in ascending power of x. [03]

(d) Find the Fourier cosine series for the function [03]

$$f(t) = \begin{cases} 1, & \text{for } 0 < t < \frac{\pi}{2}, \\ 0, & \text{for } \frac{\pi}{2} < t < \pi. \end{cases}$$

OR

(d) Obtain the half-range sine series for the function $f(t) = t^2$ in the interval $(0, 3)$. [03]

(e) Using method of separation of variables, find the solution of partial differential equation $3\frac{\partial u}{\partial x} + 2\frac{\partial u}{\partial y} = 0$. [03]

OR

(e) Using method of separation of variables, find the solution of partial differential equation $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial y} + u$, where $u(x, 0) = 6e^{-3x}$. [03]

(f) Find the series solution of differential equation $\frac{d^2y}{dx^2} + y = 0$. [03]

OR

(f) Find regular singular points of the differential equation $x(x-2)^2\frac{d^2y}{dx^2} + 2(x-2)\frac{dy}{dx} + (x+3)y = 0$. [03]

(g) For Bessel function $J_n(x)$, prove that $\frac{d}{dx}(x^n J_n(x)) = x^n J_{n-1}(x)$. [02]

OR

(g) For Bessel function $J_n(x)$, prove that $\lim\limits_{x\to 0}\frac{J_n(x)}{x^n} = \frac{1}{2^n\Gamma(n+1)}$. [02]

(h) State generating function for Legendre's polynomials $P_n(x)$ and hence show that $P_n(-x) = (-1)^n P_n(x)$. [02]

SECTION II

Q-3 Answer the following questions as directed. [30]

(a)(i) Prove that $\int\limits_0^1 \left(\log\frac{1}{y}\right)^{n-1} dy = \Gamma(n)$. [02]

(ii) Evaluate the integral $\int\limits_0^1 \frac{dx}{\sqrt{1-x^4}}$. [03]

(b) Using method of least squares, find the straight line that best fits the following data [05]

x	1	2	3	4	5
y	14	27	40	55	68

(c) Find the series solution of the Hermite equation $\frac{d^2y}{dx^2} - 2x\frac{dy}{dx} + 2py = 0$, where p is constant. [05]

OR

(c) Find the series solution of the differential equation $x\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + xy = 0$. [05]

(d) Using method of separation of variable, solve the one dimensional heat equation $\frac{\partial u}{\partial t} = c^2\frac{\partial^2 u}{\partial x^2}$, where c is constant. [05]

OR

(d) A string is stretched and fastened to two points l apart. Motion is started by displacing the string into form $y = k(lx - x^2)$, where k is constant from which it is released at time $t = 0$. Find the displacement of any point on the string [05] at a distance of x from one end at time t. Find the temperature $u(x,t)$ at any time t.

(e) Find the Fourier series of function $f(t) = t$ in the interval $-1 \leq t \leq 1$. [05]

OR

(e) Find the Fourier series to represent $f(x) = \pi - t$, for $0 < t < 2\pi$. [05]

(f) State Rodrigue's formula for Legendre's polynomials $P_n(x)$. Using it find $P_0(x), P_1(x)$ and $P_2(x)$. Also express function $f(x) = 5x^2 - 7x - 11$ in terms of Legendre polynomials. [05]

OR

(f) Write the expression of Bessel function $J_n(x)$. Prove that $xJ_n' = nJ_n - xJ_{n+1}$. [05]

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